Simple Array Program

// Q1. Declare and Display an Array

```
#include <stdio.h>


void main() {
   printf("\nQ1. Declare and Display an Array \n\n");
   int arr[] = {10, 20, 30, 40, 50};
   int size = 5;


   printf("Array elements: ");
   for(int i = 0; i < size; i++) {
      printf("%d ", arr[i]);
   }
   printf("\n");
}
```

//Q2.Take 5 Numbers and Display

```
#include<stdio.h>
void main() {
   printf("\nQ2. Take 5 Numbers and Display\n\n");
   int arr[5];


   printf("Enter 5 numbers: ");
   for(int i = 0; i < 5; i++) {
      scanf("%d", &arr[i]);
   }
```

```
    printf("You entered: ");
    for(int i = 0; i < 5; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
```

// Q3. Find the sum of all elements

```
#include<stdio.h>
void main() {
    printf("\nQ3. Sum of Array Elements\n\n");
    int arr[] = {10, 20, 30};
    int size = 3;
    int sum = 0;

    for(int i = 0; i < size; i++) {
        sum = sum + arr[i];
    }

    printf("Array: {10, 20, 30}\n");
    printf("Sum: %d\n", sum);
}
```

// Q4: Calculate average of array elements

```
#include<stdio.h>
void main() {
    printf("\n 4. Average of Array Elements\n\n");
```

```
    int arr[] = {10, 20, 30};
    int size = 3;
    int sum = 0;
    float average;

    for(int i = 0; i < size; i++) {
        sum = sum + arr[i];
    }

    average = sum / (float)size;

    printf("Array: {10, 20, 30}\n");
    printf("Average: %.2f\n", average);
}
```

// Q5: Count even and odd numbers

```
#include<stdio.h>
void main() {
    printf("\nQ5. Count Even and Odd Numbers \n\n");
    int arr[] = {1, 2, 3, 4};
    int size = 4;
    int even_count = 0, odd_count = 0;

    for(int i = 0; i < size; i++) {
        if(arr[i] % 2 == 0) {
            even_count++;
        } else {
            odd_count++;
```

```
    }
  }

  printf("Array: {1, 2, 3, 4}\n");
  printf("Even numbers: %d\n", even_count);
  printf("Odd numbers: %d\n", odd_count);
}
```

// Q6: Find the largest element

```
#include<stdio.h>
void main() {
  printf("\nQ6: Find Largest Element \n\n");
  int arr[] = {12, 45, 23};
  int size = 3;
  int largest = arr[0];

  for(int i = 1; i < size; i++) {
    if(arr[i] > largest) {
      largest = arr[i];
    }
  }

  printf("Array: {12, 45, 23}\n");
  printf("Largest element: %d\n", largest);
}
```

// Q7: Find the smallest element

```
#include<stdio.h>
```

```
void main() {
   printf("\n Q7: Find Smallest Element ===\n");
   int arr[] = {12, 45, 23};
   int size = 3;
   int smallest = arr[0];

   for(int i = 1; i < size; i++) {
      if(arr[i] < smallest) {
         smallest = arr[i];
      }
   }

   printf("Array: {12, 45, 23}\n");
   printf("Smallest element: %d\n", smallest);
}
```

// Q8: Search for a number in array

```
#include<stdio.h>
void main() {
   printf("\n  Q8: Search for a Number in Array \n");
   int arr[] = {10, 20, 30};
   int size = 3;
   int search = 30;
   int found = 0;
   int index = -1;

   for(int i = 0; i < size; i++) {
      if(arr[i] == search) {
```

```
            found = 1;
            index = i;
            break;
        }
    }

    printf("Array: {10, 20, 30}\n");
    printf("Searching for: %d\n", search);
    if(found) {
        printf("Element found at index: %d\n", index);
    } else {
        printf("Element not found\n");
    }
}
```

// Q9: Count positive, negative and zero values

```
#include<stdio.h>
void main() {
    printf("\n Q9: Count Positive, Negative, Zero \n");
    int arr[] = {-2, 0, 5, -1};
    int size = 4;
    int positive = 0, negative = 0, zeros = 0;

    for(int i = 0; i < size; i++) {
        if(arr[i] > 0) {
            positive++;
        } else if(arr[i] < 0) {
            negative++;
```

```
        } else {
            zeros++;
        }
    }

    printf("Array: {-2, 0, 5, -1}\n");
    printf("Positive numbers: %d\n", positive);
    printf("Negative numbers: %d\n", negative);
    printf("Zeros: %d\n", zeros);
}
```

// Q10: Print array in reverse order

```
#include<stdio.h>
void program10() {
    printf("\n=== PROGRAM 10: Reverse Array ===\n");
    int arr[] = {1, 2, 3};
    int size = 3;

    printf("Original array: ");
    for(int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\nReverse array: ");
    for(int i = size - 1; i >= 0; i--) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}
```